Lctr 20: Inverse z-Transform, Pole-Zero Analysis, and Properties

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Focus: Sections 10.3–10.6 (Pages 757–774)

20.1 The Big Picture

Why This Matters

Lecture 19 taught us how to go forward: from $x[n]$ to $X(z)$. Now we need to go back. Given a system's transfer function $H(z)$ (obtained from a difference equation), how do we find $h[n]$? The method is partial fractions, exactly as with the Laplace transform. The only difference is that we work with z^{-1} instead of s, and we use the ROC (circles instead of vertical lines) to assign right-sided vs. left-sided terms.
This lecture also covers the z-transform properties table and geometric evaluation of the frequency response from pole-zero plots in the z-plane.

Roadmap:

1. Inverse z-transform via partial fractions (the practical method).
2. Worked examples: distinct real poles, repeated poles, complex poles.
3. Geometric evaluation of the frequency response from the z-plane.
4. Properties of the z-transform.
5. The initial-value theorem for causal sequences.

20.2 The Inverse z-Transform: Practical Method

20.2.1 Strategy

1. Write $X(z)$ as a proper fraction in z^{-1} (degree of numerator $\leq$ degree of denominator in z^{-1}). If not proper, perform long division first.
2. Expand into partial fractions of the form $\frac{A_i}{1-d_i z^{-1}}$.
3. Invert each term using the table pairs. Use the ROC to decide right-sided vs. left-sided.

Key Insight

The ROC determines the direction, just as in the Laplace transform. For a pole at $z = d_i$:

- ROC **outside** $|d_i|$: the term is $d_i^n u[n]$ (right-sided, causal).
- ROC **inside** $|d_i|$: the term is $-d_i^n u[-n-1]$ (left-sided, anti-causal).

20.3 Example: Distinct Real Poles, Right-Sided

Inverse z-Transform: Distinct Real Poles

Find $x[n]$ given:

$$X(z) = \frac{3 - z^{-1}}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})}, \qquad |z| > 0.5$$

Step 1: Partial fractions in z^{-1}.

$$\frac{3 - z^{-1}}{(1 - 0.5z^{-1})(1 - 0.25z^{-1})} = \frac{A}{1 - 0.5z^{-1}} + \frac{B}{1 - 0.25z^{-1}}$$

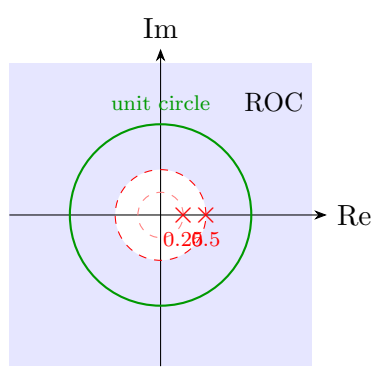
Multiply through by the denominator:

$$3 - z^{-1} = A(1 - 0.25z^{-1}) + B(1 - 0.5z^{-1})$$

Set $z^{-1} = 1/0.5 = 2$: $3 - 2 = A(1 - 0.5) \Rightarrow 1 = 0.5A \Rightarrow A = 2.$
Set $z^{-1} = 1/0.25 = 4$: $3 - 4 = B(1 - 2) \Rightarrow -1 = -B \Rightarrow B = 1.$
So: $X(z) = \dfrac{2}{1 - 0.5z^{-1}} + \dfrac{1}{1 - 0.25z^{-1}}$
Step 2: ROC is $|z| > 0.5$. Both poles ($z = 0.5$ and $z = 0.25$) are inside the ROC circle, so both terms are right-sided.
Step 3:

$$\boxed{x[n] = 2 \cdot (0.5)^n u[n] + (0.25)^n u[n]}$$

Check: $x[0] = 2 + 1 = 3$. From the initial-value theorem: $X(z)|_{z\to\infty} = 3/1 = 3$. ✓



ROC: $|z| > 0.5$ (both poles inside $\Rightarrow$ both right-sided)

20.4 Example: Mixed ROC (Two-Sided Signal)

Inverse z-Transform: Two-Sided Signal

Find $x[n]$ given:

$$X(z) = \frac{1}{(1-2z^{-1})(1-0.5z^{-1})}, \qquad 0.5 < |z| < 2$$

Step 1: Partial fractions.

$$\frac{1}{(1-2z^{-1})(1-0.5z^{-1})} = \frac{A}{1-2z^{-1}} + \frac{B}{1-0.5z^{-1}}$$

Set $z^{-1} = 0.5$: $1 = A(1-0.25) \Rightarrow A = 4/3$.
Set $z^{-1} = 2$: $1 = B(1-1) = 0$... this fails because $z^{-1} = 2$ makes $(1-2z^{-1}) = 1-4 \neq 0$.
Let us redo. Set $z^{-1} = 1/2$ to kill the $(1-2z^{-1})$ factor: $1 = A(1-0.5\cdot 0.5) = A(0.75)$, so $A = 4/3$.
Set $z^{-1} = 1/0.5 = 2$ to kill $(1-0.5z^{-1})$: $1 = B(1-2\cdot 2) = B(-3)$, so $B = -1/3$.
So:

$$X(z) = \frac{4/3}{1-2z^{-1}} + \frac{-1/3}{1-0.5z^{-1}}$$

Step 2: Assign directions from the annular ROC $0.5 < |z| < 2$:

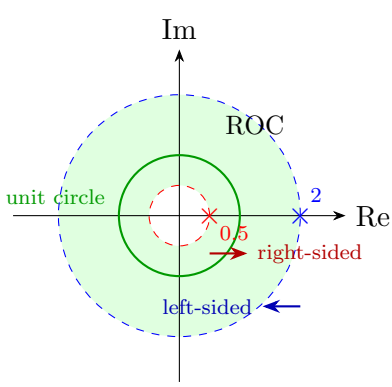
- Pole at $z = 2$ ($|z| = 2$): ROC is **inside** this pole circle $\Rightarrow$ left-sided.
- Pole at $z = 0.5$ ($|z| = 0.5$): ROC is **outside** this pole circle $\Rightarrow$ right-sided.

Step 3: Invert.

$$\frac{4/3}{1-2z^{-1}} \text{ (left-sided)} \longrightarrow -\frac{4}{3}\cdot 2^n\, u[-n-1]$$

$$\frac{-1/3}{1-0.5z^{-1}} \text{ (right-sided)} \longrightarrow -\frac{1}{3}\cdot (0.5)^n\, u[n]$$

$$\boxed{x[n] = -\frac{4}{3}\cdot 2^n\, u[-n-1] - \frac{1}{3}\cdot (0.5)^n\, u[n]}$$



ROC: $0.5 < |z| < 2$ (annular ring)

20.5 Example: Repeated Poles

Repeated Poles

Find $x[n]$ given:

$$X(z) = \frac{1}{(1-0.5z^{-1})^2}, \qquad |z| > 0.5$$

Step 1: Use the pair $na^n u[n] \mathcal{Z} \frac{az^{-1}}{(1-az^{-1})^2}$, ROC: $|z| > |a|$.
Rewrite:

$$\frac{1}{(1-0.5z^{-1})^2} = \frac{1}{(1-0.5z^{-1})} \cdot \frac{1}{(1-0.5z^{-1})}$$

We know $\frac{1}{1-0.5z^{-1}} \leftrightarrow (0.5)^n u[n]$, so by the convolution property in the z-domain (multiplication $\leftrightarrow$ convolution):

$$x[n] = (0.5)^n u[n] * (0.5)^n u[n]$$

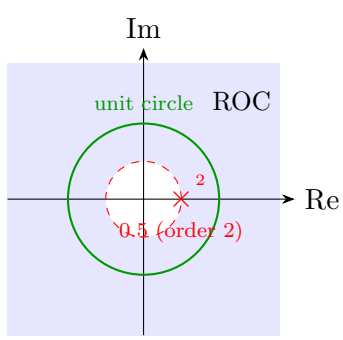
Alternatively, use the formula directly. Since $\frac{1}{(1-az^{-1})^2} = \frac{d}{d(az^{-1})} \frac{1}{1-az^{-1}} \cdot az^{-1}/az^{-1}$... this gets complicated. A cleaner approach: note that

$$\frac{1}{(1-az^{-1})^2} \;\; \mathcal{Z} \;\; (n+1)\, a^n\, u[n]$$

This can be verified by differentiating the geometric series $\sum_{n=0}^{\infty} a^n z^{-n} = \frac{1}{1-az^{-1}}$ with respect to a.

$$\boxed{x[n] = (n+1)\,(0.5)^n\, u[n]}$$

Check: $x[0] = 1 \cdot 1 = 1$, $x[1] = 2 \cdot 0.5 = 1$, $x[2] = 3 \cdot 0.25 = 0.75$. The sequence starts at 1, grows briefly, then decays. ✓



ROC: $|z| > 0.5$ (double pole, right-sided)

20.6 Example: Complex Conjugate Poles

Complex Poles: Damped Sinusoid

Find $x[n]$ given:

$$X(z) = \frac{1 - 0.8\cos(0.4\pi)\, z^{-1}}{1 - 2 \cdot 0.8\cos(0.4\pi)\, z^{-1} + 0.64\, z^{-2}}, \qquad |z| > 0.8$$

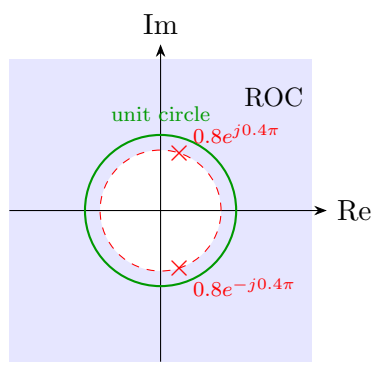
Step 1: Identify the parameters. Comparing with the table pair for $r^n \cos(\omega_0 n)\, u[n]$:

$$\frac{1 - r\cos(\omega_0)\, z^{-1}}{1 - 2r\cos(\omega_0)\, z^{-1} + r^2 z^{-2}}$$

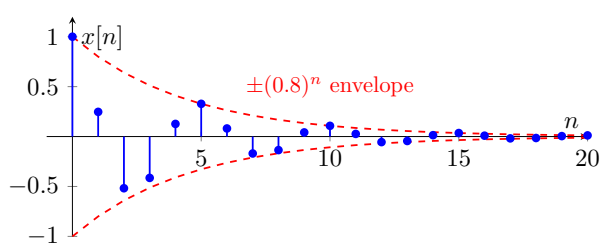
We read off: $r = 0.8$, $\omega_0 = 0.4\pi$.
The poles are at $z = 0.8\, e^{\pm j0.4\pi}$, both with $|z| = 0.8$.
Step 2: Invert.

$$\boxed{x[n] = (0.8)^n \cos(0.4\pi\, n)\, u[n]}$$

This is a discrete-time damped sinusoid with envelope $(0.8)^n$ (decays because $r < 1$) and oscillation frequency $\omega_0 = 0.4\pi$ rad/sample.



ROC: $|z| > 0.8$ (complex poles inside $\Rightarrow$ right-sided)



20.7 Geometric Evaluation of the Frequency Response

To obtain the frequency response $H(e^{j\omega})$ from the pole-zero plot, evaluate $H(z)$ on the unit circle ($z = e^{j\omega}$).

20.7.1 The Key Idea

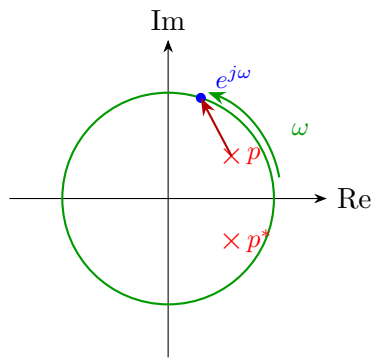
For $H(z) = K\,\dfrac{\prod_i (z - z_i)}{\prod_i (z - p_i)}$, each factor $(e^{j\omega} - z_i)$ is a vector from z_i to the point $e^{j\omega}$ on the unit circle:

$$\boxed{|H(e^{j\omega})| = |K|\,\frac{\prod_i |e^{j\omega} - z_i|}{\prod_i |e^{j\omega} - p_i|}}$$

Near a pole: the denominator vector is short $\Rightarrow$ $|H|$ is large (peak).

Near a zero: the numerator vector is short $\Rightarrow$ $|H|$ is small (notch).

20.7.2 Reading the Frequency Response from the Pole-Zero Plot



short vector from pole to $e^{j\omega}$ $\Rightarrow$ large $|H(e^{j\omega})|$

Key Insight

As ω sweeps from 0 to π (i.e., $z = e^{j\omega}$ moves along the top half of the unit circle from $z = 1$ to $z = -1$):

- At $\omega = 0$, you are at $z = 1$ (DC, the "low-frequency" end).
- At $\omega = \pi$, you are at $z = -1$ (the Nyquist frequency, the "high-frequency" end).
- Whenever $e^{j\omega}$ passes close to a pole, $|H|$ peaks.
- Whenever $e^{j\omega}$ passes close to a zero, $|H|$ dips.

Poles near the unit circle create sharp peaks; poles far from it create broad, gentle bumps. Poles on the unit circle create infinite peaks (marginally stable).

20.8 Properties of the z-Transform

Property	**Sequence**	**z-Domain / ROC**
Linearity	$ax_1[n] + bx_2[n]$	$aX_1(z) + bX_2(z)$, ROC $\supseteq R_1 \cap R_2$
Time shifting	$x[n - n_0]$	$z^{-n_0}X(z)$, ROC $= R_x$ (possibly add/remove $z = 0, \infty$)
z-domain scaling	$z_0^n x[n]$	$X(z/z_0)$, ROC $= \lvert z_0\rvert \cdot R_x$
Time reversal	$x[-n]$	$X(z^{-1})$, ROC $= 1/R_x$
Convolution	$x_1[n] * x_2[n]$	$X_1(z) \cdot X_2(z)$, ROC $\supseteq R_1 \cap R_2$
Differentiation in z	$n\,x[n]$	$-z\dfrac{d}{dz}X(z)$, ROC $= R_x$
First difference	$x[n] - x[n-1]$	$(1 - z^{-1})X(z)$
Accumulation	$\displaystyle\sum_{k=-\infty}^{n} x[k]$	$\dfrac{1}{1 - z^{-1}}X(z)$, ROC $\supseteq R_x \cap \{\lvert z\rvert > 1\}$

20.8.1 Most Important Properties

Key Insight

Convolution $\leftrightarrow$ Multiplication: $y[n] = x[n] * h[n] \;\; \mathcal{Z} \;\; Y(z) = X(z) \cdot H(z)$.
This is the foundation of DT system analysis, exactly paralleling the Laplace transform.
Time shift $\leftrightarrow$ Multiplication by z^{-n_0}: A delay of one sample corresponds to multiplying by z^{-1}. This is why z^{-1} is called the "unit delay operator" in signal processing and digital filter design.

20.8.2 Worked Example: Time Shifting

Time Shifting in the z-Domain

Given $x[n] = (0.5)^n u[n] \mathcal{Z} \frac{1}{1-0.5z^{-1}}$, $|z| > 0.5$.
Find the z-transform of $y[n] = x[n-3] = (0.5)^{n-3}u[n-3]$.
By the time-shift property:

$$\boxed{Y(z) = z^{-3} \cdot \frac{1}{1 - 0.5z^{-1}} = \frac{z^{-3}}{1 - 0.5z^{-1}}, \quad |z| > 0.5}$$

The delay of 3 samples introduces the factor z^{-3}, which adds 3 zeros at $z = 0$.

20.8.3 The Initial-Value Theorem

Initial-Value Theorem for Causal Sequences

If $x[n] = 0$ for $n < 0$:

$$\boxed{x[0] = \lim_{z\to\infty} X(z)}$$

This is simpler than the Laplace version (no multiplication by s). For a proper rational $X(z)$ in z^{-1}, the limit as $z \to \infty$ just sets all z^{-1} terms to zero.

20.8.4 The Final-Value Theorem

Final-Value Theorem for Causal Sequences

If $x[n]$ has a finite limit as $n \to \infty$ (i.e., all poles of $(1 - z^{-1})X(z)$ are strictly inside the unit circle):

$$\boxed{\lim_{n\to\infty} x[n] = \lim_{z\to 1} (1 - z^{-1})\, X(z)}$$

Warning

The final-value theorem is only valid if $\lim_{n\to\infty} x[n]$ actually exists as a finite number. If $x[n]$ oscillates (poles on the unit circle at $z \neq 1$) or grows (poles outside the unit circle), the theorem gives a meaningless result. Always check that $(1-z^{-1})X(z)$ has all its poles strictly inside the unit circle before applying it.
Compare with the Laplace version $x(\infty) = \lim_{s\to 0} s\,X(s)$: the factor $(1 - z^{-1})$ in the z-domain plays the same role as s in the s-domain. Both "cancel" the pole that a step-like signal creates (at $z = 1$ or $s = 0$, respectively).

Initial and Final Value Theorems

Given: $X(z) = \dfrac{5}{(1 - z^{-1})(1 - 0.6z^{-1})}$, ROC: $|z| > 1$.
Initial value: $x[0] = \lim_{z\to\infty} X(z) = \frac{5}{(1)(1)} = 5$.
Final value check: $(1 - z^{-1})X(z) = \frac{5}{1-0.6z^{-1}}$, which has a single pole at $z = 0.6$ (inside the unit circle). The theorem applies. ✓
$x[\infty] = \lim_{z\to 1}(1 - z^{-1})X(z) = \lim_{z\to 1} \frac{5}{1-0.6z^{-1}} = \frac{5}{1-0.6} = \frac{5}{0.4} = 12.5$
So $x[n]$ starts at 5 and settles to 12.5 as $n \to \infty$.

20.9 Summary

1. **Partial fractions in** z^{-1} is the practical inversion method. Decompose into $\frac{A}{1-dz^{-1}}$ terms and look up each pair.

2. The ROC determines right-sided vs. left-sided: ROC outside the pole $\Rightarrow d^n u[n]$; ROC inside $\Rightarrow -d^n u[-n-1]$.

3. **Complex poles** at $z = re^{\pm j\omega_0}$ produce $r^n \cos(\omega_0 n)$ or $r^n \sin(\omega_0 n)$ responses.

4. The **frequency response** is read geometrically from the pole-zero plot by sweeping $z = e^{j\omega}$ around the unit circle: short vectors to poles $\Rightarrow$ peaks; short vectors to zeros $\Rightarrow$ notches.

5. The **convolution property** turns DT convolution sums into algebraic multiplication in the z-domain; the **time-shift property** makes z^{-1} the unit delay operator.

20.10 Common Mistakes to Avoid

1. **Doing partial fractions in positive powers of z:** Always work in z^{-1}. The standard pairs are expressed in z^{-1} form. If you do partial fractions in z, you will need extra algebra to convert back.

2. **Wrong direction assignment:** ROC outside the pole $\Rightarrow$ right-sided. ROC inside $\Rightarrow$ left-sided. ("Outside" means $|z|$ is *larger* than the pole magnitude.)

3. **Forgetting the negative sign for left-sided terms:** The pair is $-a^n u[-n-1]$, not $a^n u[-n-1]$.

4. **Improper fractions:** If the degree of numerator in z^{-1} equals or exceeds the degree of denominator, you must do long division first to extract a polynomial in z^{-1} (which inverts to δ functions) before applying partial fractions.

5. **Sweeping ω in the wrong direction:** $\omega = 0$ corresponds to $z = 1$ (not $z = 0$). $\omega = \pi$ corresponds to $z = -1$.

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